
1. Indexer

bin/magento indexer:reindexer

two ways ->

- using cmd
- using admin

Main two modes->

- 1.Update On Save
- 2.Update By Schedule

Command	Description
php bin/magento indexer:reindex	Initiates the reindex of all indexers

php bin/magento indexer:reindex [index_type]	Reindexes a specific index type
php bin/magento indexer:info	Shows the list of all available indexers
php bin/magento indexer:status	Displays the indexers' status
php bin/magento indexer:show-mode	Shows the indexers' mode
php bin/magento indexer:set-mode	Sets the Indexer mode
php bin/magento indexer:reset	Sets the indexer status to <i>Invalid</i>

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-- > updating the information in the index tables to keep it up-to-date

1. indexer:reindex

- rebuilt (reset) data , update the indexed data
- design config , customer grid , category product , product category , catalog rule product, product eav index , stock index , product price , catalog product rule , catalog search index...
- product EAV index -> products in catalog & attribute
- stock index ->

-- > Practical Use Cases for Reindexing :-

- Adding or Updating Products
- Price Changes
- Category Changes
- Design Changes

2. indexer:info

- key and value
- key – data base table name's
- value - title

3. indexer:status

- display indexer status

4.indexer:show-mode

- title & Update on Save
- show thw mode of all indexer

5. indexer:set-mode {realtime|schedule} [indexer]

ex:- bin/magento indexer:set-mode schedule catalog_category_product
catalog_product_category

2. virtual type

- use in di.xml
- use in grid for connection
- two attribute
- > name : set v_type name as value or (custom file path)
- > type : (vender file path)connect with db table

->argument : database table name & resource path

ex1:-

```
<virtualType name="ourVirtualTypeName"
type="Mageplaza\HelloWorld\Model\Virtualtype"> </virtualType>
<type name="Mageplaza\HelloWorld\Model\Example">
    <arguments>
        <argument name="the_object"
            xsi:type="object">ourVirtualTypeName</argument>
    </arguments>
</type>
```

ex2:-

```
<virtualType
name="Magento\Catalog\Ui\DataProvider\Product\ProductCollectionFactor
y"
type="Magento\Catalog\Model\ResourceModel\Product\CollectionFactory">
    <arguments>
        <argument name="instanceName"
            xsi:type="string">\Magento\Catalog\Ui\
DataProvider\Product\ProductCollection</argument>
    </arguments>
</virtualType>
```

--> what is Virtual Type?

- Alias or Nickname – nickname for a real class or setup in magento(argument

name="")

- Indirect Use -

-->why it use...

- Flexible , reuse ,simple

3. difference between object and registry

- - > **Object :**

- use classes

- - > **Registry :**

- use globl , set get data un-set data

4. why we use upgrade command

-----> This command is used

to apply changes to the database schema, data, and to update the module version.

1. Apply database schema changes
2. Run data patches
3. Update module version
4. Recompile dependency Injection Configurations
5. Execute Upgrade Scripts
6. C:C and generated files

When to Use `setup:upgrade`

- change to module's database schema or data
- installing a new module an existing module
- change version of module
- after deploying code that includes database or data change.
- change or add something in construcoter
- vendor ma thi kai add kraia tyare

- c:c ,schema update , file clean , module ,diretory

- cleate system : static content file (pub/static) and generated

- module install

- create database schema

- create attributes

5. why we use S:D:C command

- s:d:c – setup di compile

- missing class will be auto generated

- di code ne compile kria ana mate use thay

- plugin , event , constructor add ,

1.Dependency Injection Compilation

2. Generating Proxies and Factories

3. Code Generation

4. Performance Optimization

5. Consistency and Error Detection

When to Use `setup:di:compile`

- After installing or updating modules.
- After making changes to di configuration

repo , appliction intercepor ,interception catch , plugin , area config

6. why we use S:S:D command

---> `setup : static-content:deploy`

js/css/html add kria tyre...

Deploys static view files

7. OpenSearch v/s Elastic Search

- Both open-source search and analytics engines
 - - >Elastic Search
 - additional paid features like security , machine learning , support options etc...
 - widely popular due to its scalability, flexibility, and powerful search capabilities.
 - - > Open Search
 - build an open source community around search and analytic
 - cost savings